CHAPTER 1

Sequences and induction

S2, S4, S5, S7, S10, S11, S13

Objectives

Calculate terms and a sum; write up a proof by induction; establish monotonicity and bounds; use an auxiliary sequence.

Course notes

Defining a sequence and calculating

A real sequence is a function defined on $\mathbb{N}$, or on the integers from some index n_0 onwards. We write u_n for its term of index n. An explicit formula gives u_n directly; a recurrence relation also requires an initial term. For example, $u_0 = 2$ and $u_{n+1} = 3u_n - 1$ give $u_1 = 5$, $u_2 = 14$.

Reference sequences

If $u_{n+1} = u_n + r$, then $u_n = u_p + (n-p)r$. If $u_{n+1} = qu_n$, then $u_n = u_p q^{n-p}$ for $n \geq p$.

$$\sum_{k=p}^{n} u_k = (n-p+1)\frac{u_p + u_n}{2} \quad \text{(arithmetic)}, \qquad \sum_{k=p}^{n} u_k = u_p \frac{1-q^{n-p+1}}{1-q} \quad (q \neq 1).$$

When $q = 1$, the second sum equals $(n-p+1)u_p$.

Proof by induction

To prove $P(n)$ for every integer $n \geq n_0$: check $P(n_0)$; fix an arbitrary integer $n \geq n_0$ and assume $P(n)$ is true; deduce $P(n+1)$; conclude by the principle of induction. The assumption is not the conclusion: it can only be used at the fixed index.

Fully worked example. Let $u_0 = 1$, $u_{n+1} = 2u_n + 3$. For $n \in \mathbb{N}$, let $P(n)$ be the statement "$u_n = 4 \times 2^n - 3$".

Base case. $4 \times 2^0 - 3 = 1 = u_0$; $P(0)$ is true.

Inductive step. Fix $n \in \mathbb{N}$ and assume $P(n)$ is true. By the definition of the sequence, then by this assumption,

$$\begin{aligned} u_{n+1} &= 2u_n + 3 \\ &= 2(4 \times 2^n - 3) + 3 \\ &= 8 \times 2^n - 3 = 4 \times 2^{n+1} - 3. \end{aligned}$$

This is $P(n+1)$: the property is inherited.

Conclusion. By the principle of induction, $u_n = 4 \times 2^n - 3$ for every $n \in \mathbb{N}$.

Proof: sum of the integers. Let $P(n)$ be the statement "$\sum_{k=0}^{n} k = \frac{n(n+1)}{2}$". For $n = 0$, both sides equal 0: $P(0)$ is true. Fix $n \geq 0$ and assume $P(n)$ is true. Then

$$\begin{aligned}\sum_{k=0}^{n+1} k &= \sum_{k=0}^{n} k + (n+1)\\ &= \frac{n(n+1)}{2} + (n+1)\\ &= \frac{(n+1)(n+2)}{2}.\end{aligned}$$

We have established $P(n+1)$. By induction, the formula is true for every $n \in \mathbb{N}$.

Global behaviour

A sequence is increasing if $u_{n+1} \geq u_n$ for every index, decreasing if the inequality is reversed. It is bounded above if there is a real number M such that $u_n \leq M$ for every n; bounded below if there is m such that $u_n \geq m$; bounded if it is both bounded below and bounded above. To study monotonicity, one can study the sign of $u_{n+1} - u_n$. If all the terms are strictly positive, one can compare $\frac{u_{n+1}}{u_n}$ with 1. Studying a function f such that $u_n = f(n)$ is another possibility.

Arithmetico-geometric sequence

For $u_{n+1} = au_n + b$, $a \neq 1$, solve $\ell = a\ell + b$, then set $v_n = u_n - \ell$. This gives $v_{n+1} = av_n$, hence

$$u_n = \ell + (u_0 - \ell)a^n, \qquad \ell = \frac{b}{1-a}.$$

If $a = 1$, the sequence is arithmetic: do not divide by $1 - a$.

Example. If $u_0 = 10$ and $u_{n+1} = 0.8u_n + 6$, then $\ell = 30$ and $u_n = 30 - 20(0.8)^n$. We have $10 \leq u_n < 30$ and $u_{n+1} - u_n = 4(0.8)^n > 0$.

Pitfall. Checking a property on a hundred terms does not prove it. An increasing sequence is not necessarily positive; a bounded sequence is not necessarily monotonic.

Research activity

A reservoir topped up every day — 35 min.

A reservoir initially contains 100 litres of water. Each day, 20 % of the volume present is drawn off, then 30 litres are added. For every natural number n, let u_n be the volume of water, in litres, after these two operations have been carried out n times. Thus $u_0 = 100$.

1. Calculate u_1 and u_2, then express u_{n+1} in terms of u_n.
2. Which volume would remain unchanged after one day? Prove by induction that, for every $n \in \mathbb{N}$, $100 \leq u_n < 150$.
3. For every $n \in \mathbb{N}$, let $v_n = u_n - 150$. Show that (v_n) is geometric. Deduce the expressions of v_n and u_n in terms of n.
4. Write a Python function with no argument that returns the first integer n for which $u_n > 145$. Run the program and interpret the result.
5. A student has calculated the first hundred terms and notes that they are all less than 150. Is this observation enough to prove that all the terms are? Justify.

Progressive exercises

E1 · Practice. Consider the sequence (u_n) defined, for every natural number n, by $u_n = 5 - 3n$. Calculate u_4, then the sum $\sum_{k=0}^{20} u_k$.

E2 · Practice. The sequence (v_n) is defined by $v_0 = 3$ and, for every natural number n, $v_{n+1} = 2v_n$. Express v_n in terms of n, then calculate $\sum_{k=0}^{n} v_k$.

E3 · Practice. Prove by induction that, for every natural number n, $3^n \geq 2n + 1$.

E4 · Practice. Consider the sequence (u_n) defined on $\mathbb{N}$ by $u_n = \frac{n+1}{n+2}$. Prove that it is increasing and give bounds valid for all its terms.

E5 · Practice. Prove by induction that, for every natural number n, $8^n - 1$ is divisible by 7. Do not use congruences.

Textbook or past-paper work

Barbazo, p. 28, exercises 22 and 25. In class, in S5: exercise 22, to translate properties of sequences using a quantifier. At home after S11: the whole of exercise 25, to combine induction, upper bounds and monotonicity. Write the base case, the inductive step and the conclusion separately.

Homework

Homework 1 — A sum and a threshold — 45 min, after S11.

Consider the sequence (u_n) defined by $u_0 = 2$ and, for every $n \in \mathbb{N}$, $u_{n+1} = \frac{1}{2}u_n + 3$.

1. Prove by induction that $2 \leq u_n < 6$ for every $n \in \mathbb{N}$. Deduce the direction of variation of (u_n).
2. Let $v_n = u_n - 6$ for every $n \in \mathbb{N}$. Show that (v_n) is geometric, then express u_n in terms of n.
3. For every $n \in \mathbb{N}$, let $S_n = \sum_{k=0}^{n} u_k$. Express S_n in terms of n.
4. Using the explicit formula and powers of 2, determine the smallest integer n such that $u_n > 5.99$. Check that the previous integer does not work.
5. Express $6 - \frac{S_n}{n+1}$ and justify that this difference is strictly positive. Interpret the result by considering the mean of the first $n + 1$ terms; no limit calculation is required.

Teaching notes and solutions

Teacher notes

In S2, immediately distinguish the index from the value of the term. In S4, have a loop executed by hand before Python. In S5 and S7, get students to say aloud "we assume at index n, we prove at index $n+1$". The sum and the divisibility come in S10, after Test 1: do not assess them in the first test. In S11, the search for the fixed point is an algebraic technique; the convergence theorem will be established in the next chapter. For weaker students, supply the property $P(n)$ and let them write the three steps; for the strongest, remove the hint about an auxiliary sequence.

Solutions to the activity

1. $u_0 = 100$ L. After one day, $0.8u_0$ L remain, then 30 L are added: $u_1 = 110$ L. Similarly, $u_2 = 0.8 \times 110 + 30 = 118$ L. For every $n \geq 0$, $u_{n+1} = 0.8u_n + 30$.

2. The equilibrium volume L would have to satisfy $L = 0.8L + 30$, that is $L = 150$ L. Let us prove $P(n)$: "$100 \leq u_n < 150$". At index 0, $u_0 = 100 \in [100, 150[$. Fix $n \geq 0$ and assume $P(n)$ is true. Since $0.8 > 0$,
$$110 = 0.8 \times 100 + 30 \leq 0.8u_n + 30 < 0.8 \times 150 + 30 = 150.$$
Hence $100 \leq u_{n+1} < 150$ and $P(n+1)$ is true. By induction, the bounds hold for every $n \in \mathbb{N}$.

3. $v_{n+1} = u_{n+1} - 150 = 0.8u_n - 120 = 0.8(u_n - 150) = 0.8v_n$. The sequence (v_n) is geometric, with common ratio 0.8 and first term $v_0 = -50$. Thus $v_n = -50(0.8)^n$, then $u_n = 150 - 50(0.8)^n$.

4.
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def premier_jour():
    n, u = 0, 100
    while u <= 145:
        u = 0.8*u+30
        n += 1
    return n
```
The loop returns 11. Indeed, $u_{10} \simeq 144.63 < 145$ and $u_{11} \simeq 145.71 > 145$. Moreover, $u_{n+1} - u_n = 10(0.8)^n > 0$: the threshold was not crossed earlier. Calculating these two values guarantees the loop stops here, without yet requiring a convergence theorem.

5. The simulation only calculates a finite number of terms. The inductive step, together with the base case, establishes the bounds for all integers; it is this that rules out any excess over 150 L.

E1. $u_4 = 5 - 3 \times 4 = -7$. The sequence is arithmetic with common difference -3 and $u_{20} = -55$. The sum has 21 terms: $\sum_{k=0}^{20} u_k = \frac{21(u_0+u_{20})}{2} = \frac{21(5-55)}{2} = -525$.

E2. The sequence is geometric with common ratio 2 and first term $v_0 = 3$. Hence $v_n = 3 \times 2^n$. The sum has $n+1$ terms and equals $\frac{3(1-2^{n+1})}{1-2} = 3(2^{n+1} - 1)$.

E3. Let $P(n)$ be the statement "$3^n \geq 2n + 1$". At index 0, $3^0 = 1 = 2 \times 0 + 1$. Fix $n \geq 0$ and assume $P(n)$ is true. Multiplying by $3 > 0$, $3^{n+1} \geq 6n + 3$. Now $6n + 3 - (2n + 3) = 4n \geq 0$. Thus $3^{n+1} \geq 2(n+1) + 1$, which is $P(n+1)$. By induction, $P(n)$ is true for every $n \in \mathbb{N}$.

E4. For $n \geq 0$, $u_n = 1 - \frac{1}{n+2}$. We have $0 < \frac{1}{n+2} \leq \frac{1}{2}$, hence $\frac{1}{2} \leq u_n < 1$. Moreover,
$$u_{n+1} - u_n = \frac{1}{n+2} - \frac{1}{n+3} = \frac{1}{(n+2)(n+3)} > 0.$$
The sequence is strictly increasing, bounded below by $\frac{1}{2}$ and bounded above by 1.

E5. Let $P(n)$ be the statement "there exists an integer k such that $8^n - 1 = 7k$". For $n = 0$, $8^0 - 1 = 0 = 7 \times 0$: $P(0)$ is true. Fix $n \geq 0$ and assume $P(n)$ is true. There therefore exists an integer k such that $8^n - 1 = 7k$. Then

$$8^{n+1} - 1 = 8(8^n - 1) + 7 = 8 \times 7k + 7 = 7(8k + 1).$$

Since $8k + 1$ is an integer, $P(n+1)$ is true. By induction, 7 divides $8^n - 1$ for every $n \geq 0$.

Solutions to Homework 1

1. Let $P(n)$ be the statement "$2 \leq u_n < 6$". At index 0, $u_0 = 2$ satisfies the bounds. Fix $n \geq 0$ and assume $P(n)$ is true. Then $4 \leq \frac{u_n}{2} + 3 < 6$; in particular $2 \leq u_{n+1} < 6$. By induction, $P(n)$ is true at every index. We deduce $u_{n+1} - u_n = 3 - \frac{u_n}{2} = \frac{6-u_n}{2} > 0$. The sequence is strictly increasing.

2. The equation $L = \frac{L}{2} + 3$ gives $L = 6$. Let $v_n = u_n - 6$. Then $v_{n+1} = \frac{u_n}{2} - 3 = \frac{v_n}{2}$ and $v_0 = -4$. Hence $v_n = -\frac{4}{2^n}$ and $u_n = 6 - \frac{4}{2^n}$.

3. By linearity of sums and the geometric formula,

$$\begin{aligned} S_n &= 6(n+1) - 4\sum_{k=0}^{n} (\frac{1}{2})^k \\ &= 6(n+1) - 8(1 - 2^{-(n+1)}). \end{aligned}$$

4. $u_n > 5.99$ is equivalent to $\frac{4}{2^n} < 0.01$, that is $2^n > 400$. Since $2^8 = 256 \leq 400 < 512 = 2^9$ and (2^n) is increasing, the smallest index is 9.

5. The mean is less than 6 and its distance from 6 is

$$6 - \frac{S_n}{n+1} = \frac{8(1 - 2^{-(n+1)})}{n+1}, \qquad 0 < 6 - \frac{S_n}{n+1} < \frac{8}{n+1}.$$

   To make this distance less than a precision $\varepsilon > 0$, it suffices to take $n + 1 > \frac{8}{\varepsilon}$. This is a quantitative control of the distance, which does not yet require a theorem on limits.

Solutions to the textbook or past-paper work

Exercise 22. The four translations are, for every $n \in \mathbb{N}$: $u_{n+1} \geq u_n$; $v_n \geq 5$; $w_n \leq -1$; $t_n > 0$. Monotonicity compares two consecutive terms; a bound compares each term with one and the same number.

Exercise 25. We have $u_0 = -5$ and $u_{n+1} = \frac{2}{3}u_n - 1$.

Base case. $u_0 = -5 \leq -3$: the property is true at index 0.

Inductive step. Fix $n \in \mathbb{N}$ and assume $u_n \leq -3$. Multiplication by $\frac{2}{3} > 0$ preserves order, so $\frac{2}{3}u_n \leq -2$. Thus $u_{n+1} = \frac{2}{3}u_n - 1 \leq -3$.

Conclusion. By induction, $u_n \leq -3$ for every $n \in \mathbb{N}$. Next, $u_{n+1} - u_n = \frac{2}{3}u_n - 1 - u_n = -\frac{1}{3}u_n - 1$. The previous bound gives $-\frac{1}{3}u_n \geq 1$, hence $u_{n+1} - u_n \geq 0$. The sequence is increasing. The reversal of the inequality here comes from the negative factor $-\frac{1}{3}$.